Fiscal Year:	FY 2012	**Task Last Updated:**	FY 04/20/2012
PI Name:	Paul, Anna-Lisa Ph.D.		
Project Title:	The biological impact of spaceflight and extraterrestrial environments on molecular signaling and gene expression in plants		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	(352) 273-4855
Organization Name:	University of Florida		
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City:	Gainesville	**State:**	FL
Zip Code:	32611-0690	**Congressional District:**	3
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	06/09/2009	**End Date:**	06/08/2013
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Extended to 6/8/2013 per PI/NSSC info (Ed., 4/20/2012)		
Key Personnel Changes/Previous PI:	Agata Zupanska - Postdoctoral Research Associate ; Justin Goodwyn - Undergraduate Research Student ; Fiona Denison - Postdoctoral Research Associate ; Claire Amalfitano - Biological Scientist		
COI Name (Institution):	Ferl, Robert (University of Florida)		
Grant/Contract No.:	NNX09AL96G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The overall goal of our program and this proposal is to understand the fundamental molecular responses of plants to spaceflight environments, with particular emphasis on the changes in gene expression and signal transduction that accompany spaceflight experience. We have used genetically engineered plants and molecular tools to monitor plant responses to spaceflight and space-related analog environments and have developed specific hypotheses regarding plant signaling responses that are affected by spaceflight. The focus of this proposal is to build on these extensive insights to develop the next generation of biological assays and hardware concepts to address these hypotheses in ground and analog studies. In the process, these new biological and analytical tools will be integrated into functional concepts than can be readied for future deployment in a range of spaceflight-related opportunities. The biology will include new biosensors to test hypotheses regarding the mechanisms by which molecular signals are conveyed in response to spaceflight and analog environments. Some biosensors will be linked to Green Fluorescent Protein variants to facilitate multiplexed telemetric data collection. The analytical techniques will make use of current spaceflight imaging technologies and explore advanced imaging concepts. This combination of biosensors and imaging will enable robust robotic experiments with flexible future deployment options, enabling experiments that address science needs to survey molecular responses in a variety space venues and situations. The expected significance of this project is that it will advance the fundamental understanding of spaceflight molecular signaling in spaceflight-relevant analogs and then provide the means to transition these studies for tests in bona fide spaceflight environments as opportunities develop. Thus the objectives of this proposal are completely aligned with the request of the NRA to investigate the nature of cellular receptors, signal transduction pathways and changes in gene expression in response to spaceflight-relevant environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The expected significance of this project is that it will advance the fundamental understanding of molecular signaling in spaceflight-relevant analogs and then provide the means to transition these studies for tests in bona fide spaceflight environments as opportunities develop. Thus the objectives of this proposal are completely aligned with the request of the NRA to investigate the nature of cellular receptors, signal transduction pathways and changes in gene expression in response to spaceflight-relevant environments.
Task Progress:	The third year of the grant took advantage of the returned Flight materials from the PIs' two Flight Experiments: APEX-TAGES and BRIC-16. FSB support enabled an additional level of analyses of these samples, and facilitated supplemental ground studies that continue to contribute to our understanding of the response of biology to spaceflight and novel environments. The data analyses led to the design and support of several analog studies initiated last year: 1) clinorotation and gravity vector disruption, 2) hypergravity tests with a 2-3g centrifuge, and 3) a parabolic flight campaign opportunity (FO award). The concomitant ground and flight support represented by these grants enables us to combine the processes of flight biology preparation, flight data analysis, and hypothesis development and testing for unprecedented insight into biological responses to these novel environments. The execution of this grant was (and continues to be) the ideal example of what NASA intended as the role of Fundamental Space Biology Ground-Based Research: NRA-01-OBPR-06 "…to expand our understanding of fundamental biological processes and the mechanisms by which these processes sense, respond, adapt to, and evolve in the space environment…and in particular… the role of the genome and cellular structures in sensing and responding to gravitational force."
Bibliography Type:	Description: (Last Updated: 11/20/2025)
Articles in Peer-reviewed Journals	Paul A-L, Zupanska A, Ostrow DT, Zhang Y, Sun Y, Li J-L, Shanker S, Farmerie WG, Amalfitano CE, Ferl RJ. "Spaceflight transcriptomes: unique responses to a novel environment." Astrobiology. 2012 Jan;12(1):40-56. Epub 2012 Jan 5. PubMed PMID: 22221117 ; http://dx.doi.org/10.1089/ast.2011.0696 , Jan-2012
Articles in Peer-reviewed Journals	Paul A-L, Manak MS, Mayfield JD, Reyes MF, Gurley WB, Ferl RJ. "Parabolic flight induces changes in gene expression patterns in Arabidopsis thaliana." Astrobiology. 2011 Oct;11(8):743-58. Epub 2011 Oct 4. PubMed PMID: 21970703 ; http://dx.doi.org/10.1089/ast.2011.0659 , Oct-2011
Articles in Peer-reviewed Journals	Ferl RJ, Zupanska A, Spinale A, Reed D, Manning-Roach S, Guerra G, Cox DR, Paul A-L. "The performance of KSC Fixation Tubes with RNALater for orbital experiments: A case study in ISS operations for molecular biology." Advances in Space Research 2011 Jul;48(1):199-206. http://dx.doi.org/10.1016/j.asr.2011.03.002 , Jul-2011
Articles in Peer-reviewed Journals	Paul A-L, Ferl RJ. "Using Green Fluorescent Protein (GFP) reporter genes in RNAlater™ fixed tissue." Gravitational and Space Biology 2011 Sep;25(1):40-3. http://gravitationalandspacebiology.org/index.php/journal/article/view/530 , Sep-2011
Articles in Peer-reviewed Journals	Denison FC, Paul A-L, Zupanska AK, Ferl RJ. "14-3-3 proteins in plant physiology." Semin Cell Dev Biol. 2011 Sep;22(7):720-7. Epub 2011 Aug 31. PubMed PMID: 21907297 , Sep-2011